Alcovia Assignment – Product, Systems & Analytics Lead

Oct 16, 2026 · @juhi

Team and approach

In the first 8 weeks from 1 Nov, the team fixes app adoption first, ships the GTM tracker by week 3, and has the student dashboard live by week 8. The delivery process from Problem 3 runs all of it.

Person	Owns	Does not own
Me (Product, Systems & Analytics Lead)	What we build and why, success metrics, specs, data model, weekly review, talking to students, parents and advisors	Writing production code
Full-stack engineer	App changes, Mixpanel tracking, health and wearable integrations, student dashboard views	Deciding priorities
Process / systems designer	ClickUp structure, n8n automations, GTM dashboard (no-code), Smallest.ai call flows, advisor and sales workflows, SOPs and training	App code

One plan, checked against capacity

The constraint is the single engineer. App changes, GTM data syncs and wearable integrations cannot all be built at once. So the GTM dashboard is built without code by the process designer (n8n → Google Sheets → Looker Studio), and the engineer stays on the app.

Weeks	Me	Full-stack engineer	Process designer
1 (1–7 Nov)	Metric definitions for all three problems; 6 student interviews; GTM targets agreed with founders	Fix Mixpanel tracking; measure load times; 3-day feasibility check on focus mode and health data	ClickUp lead statuses matched to funnel stages; Razorpay and Meta connected to Sheets via n8n

Weeks	Me	Full-stack engineer	Process designer
2–3 (8–21 Nov)	App specs; check every GTM number against its source	New onboarding and Today card	GTM dashboard live (end of week 3); Smallest.ai reminder and renewal calls
4–5 (22 Nov–5 Dec)	Test app changes with 5 students; design student dashboard with 2 advisors	Focus → chapter → Realm reward; 8pm quiz in app; pod ranking	Clean up student data in ClickUp; rating prompts, facilitator forms, n8n nudges
6–7 (6–19 Dec)	Read app results; decide scale, iterate or kill	Health Connect and HealthKit reading; student dashboard views	Advisor training; parent weekly summary
8 (20–26 Dec)	Retro; plan January	Fixes; faster Study and Tasks tabs	SOPs and checklists handed over

Not in the first 8 weeks: the phone-locking focus mode. Week 1 checks feasibility; the build starts in January if approved (see Problem 4).

Problem 1 – App adoption (2.15 min → 30 min a day)

Kids come for the Realm game and leave because the app never turns that fun into study time. The fix is one daily loop built around a focus session, because a single 25-minute session gets a student most of the way to 30 minutes.

What the usage data says

- **Small base, already leaking:** 20 students in 45 days, 11–17 active a week, 7 gone for two weeks.
- **Fun works, study doesn't:** everyone uses Home and 13 use the Realm. Syllabus fell from 10 users to 1 this week; nobody opened Study Planner.
- **Tried, then dropped:** Focus Timer 11 tried, 5 regular. Goals had 1 completion in 45 days.
- **Some features can't work at this size:** a ranking of 20 students motivates no one, and Wellness needs a WHOOP that 1 student owns.

What I saw using the app

I recorded a first-time run on Android with the test account (2 min 20 s). Half of it went on getting into the app.

Step	What happens	Time in video	How a teen likely feels
1	Login, then a Google "save password" popup	0:00–0:08	Fine
2	Welcome screen + 4 intro screens, each listing 4 features	0:08–0:27	"16 features, I'm skimming"
3	"You're all set!" – but setup continues	0:27	"Wait, I thought I was done?"
4	Pick a goal: top ranks / body and mind / dream university	0:30–0:40	Good idea, comes too late
5	Two more intro screens (Track-Compete-Rise, Hatch Dragons)	0:42–0:47	Repeats step 2
6	Connect Google Calendar → Connect WHOOP → "Almost there"	0:48–0:58	No WHOOP, so skip
7	Home loads with a 10-step tooltip tour on top	1:00–1:27	Tapping Next without reading
8	Home: focus ring, "Daily Velocity 0.4% / 1.33%", streak, screen-time chart	1:27–1:36	"What is velocity?"
9	Study tab: loading placeholders, then subjects (Biology 7/23 chapters)	1:39–1:42	A list to update, no reward shown
10	Tasks tab: loading, then "101 of 155 done"	1:45–1:48	155 tasks feels like being behind
11	Realm: season, 3D globe, castles, dragon, 204.2K coins	1:51–2:18	The fun part, most polished

A new student taps through about 12 setup screens and 10 tooltips, roughly 22 taps, before doing anything useful.

Why students leave

1. **Onboarding explains everything before anything is used.** Features are introduced three times: intro cards, setup screens and the tooltip tour.
2. **Study and the Realm feel like two different apps.** Study and Tasks are dark lists; the Realm is a real game. Nothing on a chapter says "this earns you a castle".
3. **The language is written for advisors, not teens.** "Daily Velocity 1.33%" and "Focus Score" read like a report.

4. **Numbers overwhelm or mean nothing.** 155 tasks makes a kid feel behind; at 204.2K coins, earning 50 more feels like nothing.
5. **The real program lives outside the app.** The 8pm quiz, advisor sessions and pods give no reason to open the app daily.
6. **Small frictions add up.** About 2–3 seconds of loading on Study and Tasks on every visit.

Worth keeping: Start Focus is already the biggest button on Home, the Realm is a strong hook, and the goal-pick screen is good personalisation.

The fix: one daily loop

Open → see today's one task (set by the advisor) → start a focus session on that chapter → earn Realm progress → mark the chapter stage done → see the pod's progress.

1. **Cut onboarding to 3 screens:** pick a goal, see today's task, start the first focus session. Teach other features the first time a student reaches them.
2. **Home shows four things only:** today's task, Start Focus, the 8pm quiz, my pod. Everything else moves one level down.
3. **Link study to the game:** each focus session and chapter stage shows its Realm reward up front ("+1 territory").
4. **Teen language:** "1 more chapter today to stay on track" instead of velocity percentages.
5. **Show only today:** 3 tasks for today, not 155 overall. Rescale coins so one session is a visible jump.
6. **Bring the 8pm quiz into the app** – a daily appointment students already keep.
7. **Pod streaks and rankings** among 5 friends instead of all 20 students.
8. **Hide Wellness** until a wearable is connected, or allow one-tap sleep logging.
9. **Nudges through n8n:** a reminder before the student's usual study time, "your pod is ahead" alerts, a weekly parent summary.

Who does what

Weeks	Me	Full-stack engineer	Process designer
1 (1–7 Nov)	Set the metric (daily active minutes incl. focus); interview 6 students, 3 of them lapsed	Fix Mixpanel events for each loop step; measure load times	Map how advisors assign work today; design the weekly assignment flow

Weeks	Me	Full-stack engineer	Process designer
2–3 (8–21 Nov)	Write specs; test mocks with 5 students	New 3-screen onboarding and Today card	Train advisors to assign one focus task a day
4–5 (22 Nov–5 Dec)	Release to 5 students first, then all	Focus → chapter → Realm reward; 8pm quiz in app; pod ranking	n8n nudges; win-back calls to lapsed students via Smallest.ai
6–8 (6–26 Dec)	Read results weekly; decide scale, iterate or kill	Iterate on what the data shows; faster Study and Tasks	Advisor assignment and nudges become a weekly checklist

Targets

By	Avg minutes a day	Other check
30 Nov	10	15 of 20 students active weekly
31 Dec	20	Half of students do 1+ focus session a day
31 Jan	30 (stable)	Lapsed students under 3

With 20 students, an A/B test cannot give a reliable answer. Each change is judged by its before-and-after trend over 2–3 weeks plus 5 student conversations, and changes ship one at a time so their effects can be told apart.

Problem 2 – GTM tracker

One dashboard with a timeline selector answers four questions: are we on target, where do leads drop out, which ads are tired, and where should the money go next. A working version with dummy data is here: Alcovia GTM Tracker. It goes live on real data by the end of week 3.

What it shows

Block	Metrics	Flags
1. Revenue	Target vs achieved; trial vs full program; students onboarded; workshop revenue (seats × ₹3,999); program vs workshop split; unpaid invoices; trial renewals due in 30 days	Red when under 90% of target; overdue invoices by age; renewals with no call booked
2. Meta funnel	Leads → qualified → discovery call set → showed up → ambition interview → architect session → paid. Cost per qualified lead, conversion %, organic leads, average days in funnel (for those who sign up and for those who drop off, with the stage where they dropped)	Red when less than 60% of a stage carries on
3. Campaigns	Per creative: spend, CTR, frequency, results, cost per result. Landing pages: visits, drop-off, mobile load time	CTR under 1%, frequency 3+, load over 3 s, drop-off over 65%
4. Root cause	Every creative put into Scale, Refresh or Drop with the reason	Rules below

Scale, refresh or drop rules

- **Scale:** cost per result at or under target (₹2,500 per qualified lead, ₹1,200 per workshop seat) and frequency under 2.5.
- **Refresh:** frequency 3+ or CTR under 1%. New creative, same audience.
- **Drop:** cost per result over 1.6× target and CTR under 1%. The freed budget moves to the Scale group.

The targets and multipliers are starting points; we tune them after the first month of real data. All figures except the ₹3,999 workshop price are dummy values.

Where the data comes from

Data	Source	Sync
Payments, workshop seats, invoices	Razorpay	n8n, daily
Lead stages and dates	ClickUp (one status per funnel stage)	n8n, daily
Spend, CTR, frequency, leads	Meta Ads Manager	n8n, daily
Landing page load and drop-off	Google PageSpeed and site analytics	Weekly
Targets	Set monthly by founders	Manual, once a month

The one fix this needs first: ClickUp lead statuses must match the funnel stages exactly, and every stage change gets a date. The process designer owns that in week 1.

The live version is built without code: n8n pulls each source into Google Sheets every night, and Looker Studio draws the dashboard. That keeps the engineer free for the app. The dummy-data page above is the design target.

Using Smallest.ai on the weak stages

The tracker shows where the funnel leaks; Smallest.ai, which Alcovia already pays for, can act on two of those leaks automatically.

- **Discovery call show-up:** an AI reminder call 24 hours and 1 hour before each call, triggered by n8n when the ClickUp status becomes "Discovery call set". Success = show-up rate up from 75% to 85%.
- **Trial renewals:** a call 14 days before a trial ends to book the renewal conversation. Success = every renewal has a call booked 7 days ahead.

Who does what

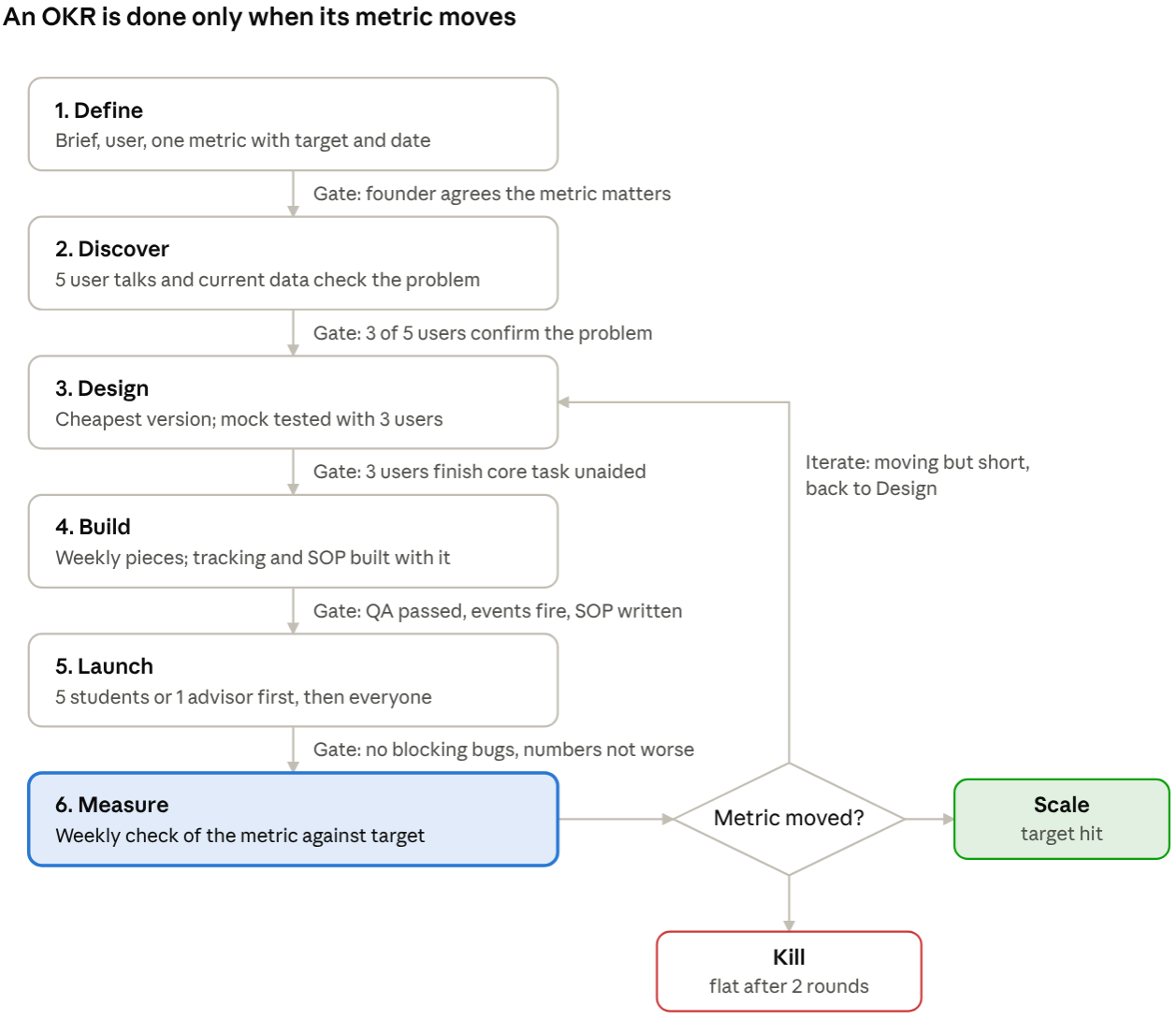
Week	Me	Engineer	Process designer
1	Lock metric definitions and targets with founders	Not needed	Rebuild ClickUp lead statuses to match the funnel; connect Razorpay and Meta to Sheets via n8n

Week	Me	Engineer	Process designer
2	Check every number against its source	Not needed	Build the Looker Studio dashboard; train sales team to update lead status the same day
3	Launch; first Friday review	Half a day to help with any API access issues	Smallest.ai reminder and renewal calls; weekly review checklist

Problem 3 – How every OKR gets delivered

Every new feature, report or system goes through the same 6 stages, and it cannot move to the next stage until it passes a gate. The rule underneath: **nothing is "done" when it ships; it is done when its success metric moves.**

The 6 stages



OKR delivery flow · 6 stages, 5 gates, 1 decision

Each stage must pass its gate before the next starts; after launch, the metric decides whether the work scales, goes back to Design, or stops.

1. **Define (1–2 days) – What problem, for whom, and how will we know it worked?**
   - Write a one-page brief: the user (teen, parent or advisor), the problem in their words, one success metric with a target and date, and what we will NOT build.
   - Example: "Teens don't study in the app. Metric: 1+ focus session a day for 50% of students by 31 Dec."
   - **Gate:** I sign off; the founder agrees the metric matters.
2. **Discover (2–5 days) – Check the problem is real before building.**
   - Talk to 5 users, look at the current data, and walk through the existing flow ourselves.
   - **Gate:** at least 3 of 5 users confirm the problem; data backs it.

3. **Design (2–5 days) – Cheapest version that can move the metric.**
   - Sketch the flow, count taps and screens, and test a clickable mock with 3 users.
   - Cognitive-load check: one main action per screen, plain teen language, no more than 3 taps to the core action.
   - **Gate:** 3 users complete the core task without help; tracking events are written into the spec.
4. **Build (1–3 weeks) – Small, weekly shippable pieces.**
   - Engineer builds; process designer builds the process side (SOP, ClickUp fields, n8n automation, advisor training).
   - Tracking is built with the feature, not after it.
   - **Gate:** QA checklist passed; events fire in Mixpanel; the SOP is written.
5. **Launch (1 week) – Small group first.**
   - Release to 5 students or 1 advisor first, then everyone. Tell users what changed and why.
   - **Gate:** no blocking bugs; early numbers are not worse than before.
6. **Measure and improve (2–4 weeks) – Did the metric move?**
   - Check the metric weekly against the target and ask users what still gets in the way.
   - Decide: **scale** (target hit), **iterate** (moving but short), or **kill** (no movement after 2 rounds).
   - **Gate:** a one-paragraph result note goes in ClickUp, so the next project learns from it.

Who owns what in each stage

Stage	Me	Engineer	Process designer
Define	Writes brief and metric	Flags tech limits	Flags process limits
Discover	Runs user talks, reads data	Pulls data	Maps current process
Design	Flow and mock	Feasibility, tracking plan	SOP and automation design
Build	Unblocks, reviews weekly	Builds feature + tracking	Builds ClickUp / n8n / SOP
Launch	Release plan, user comms	Release, monitors bugs	Trains advisors and sales

Stage	Me	Engineer	Process designer
Measure	Reads metric, decides	Fixes and iterates	Checks process is followed

The weekly rhythm that keeps it from breaking

- **Monday, 30 min – Plan:** each person picks this week's tasks from the ClickUp board; every task is linked to an OKR.
- **Daily, 10 min – Standup in ClickUp chat:** done, doing, blocked.
- **Friday, 45 min – Metrics review:** every live OKR shows its metric vs target on one screen; each gets a scale / iterate / kill call.
- **Monthly – Retro:** what slowed us down, and one process change to try.

Keeping cognitive load low for teens and parents

- One main action per screen; anything else is one level down.
- Teens get plain, short words; parents get a weekly summary, not a dashboard to dig through.
- Default values instead of forms; capture data automatically wherever possible.
- Every new feature must remove or merge something old, so the app doesn't keep growing.

Problem 4 – Student dashboard

One dashboard, fed mostly by the app and automations, shows every activity on three questions: did Alcovia do it, did the student like it, and did the student improve. It lives in the advisor web view of the existing app, goes live by week 8 (late December), and replaces the weekly docs.

A working prototype with dummy data for 20 students: Alcovia Student Tracker.

KPIs for each activity

Level	Activity	Did we do it? (quantity)	Did they like it? (quality)	Did they improve? (performance)
Day	Timetable	% of days with a timetable set	Weekly 1-tap "was today's plan realistic?"	% of planned slots completed

Level	Activity	Did we do it? (quantity)	Did they like it? (quality)	Did they improve? (performance)
Day	8pm quiz	Quizzes held / days in month	Attempt rate; quiz rating	Avg score; trend over 4 weeks
Day	Focus time	Days with 1+ session	Sessions completed vs abandoned	Minutes a day vs target
Day	Active minutes	% of students with a wearable synced	–	Active minutes vs weekly target
Day	Recovery	Sync rate	–	Avg sleep hours; recovery trend
Week	Academic sessions 1 and 2	Sessions held / planned	Session rating (1–5)	Task completion rate, deadline adherence, time-estimate accuracy
Week	Syllabus completion	Chapters tracked per subject	–	% syllabus done vs 25%-per-quarter target
Week	Rigour of completion	Chapters with each stage logged	–	Chapter readiness score (below)
Month	Workshops 1 and 2	Held / planned; attendance %	Rating; NPS	Soft-skill scores (11 skills, rated by facilitator)
Month	Pod session	Held; attendance %	Rating; "felt heard" yes/no	Peer-support actions logged
Month	Peer session	Sessions set up by advisor	Rating by both students	Topic learned (yes/no)
Month	Mentor call	Calls held / students	Rating	Next step taken within 30 days
Month	Personal branding	Shoots done; clips posted	Student happy with clip (yes/no)	Followers / reach growth
Quarter	Podcast shoot	Shot and published	Rating	Career clarity score before vs after

Level	Activity	Did we do it? (quantity)	Did they like it? (quality)	Did they improve? (performance)
Quarter	Competition prep	Students entered	Prep rating	Placed / shortlisted
Quarter	Parent-child workshop	Held; parent attendance %	Parent and teen rating	Parent NPS
Quarter	Out-of-city trip	Held; attendance %	Rating	–
Quarter	Theme (social work, startup, summer school)	Projects started	Rating	Milestones hit; applications submitted
Roadmap	Career	Roadmap made	–	Done / not done
Roadmap	Personal	Monthly milestones set	–	% milestones hit
Roadmap	Academic	Score targets set per exam	Parent and student agreed	Actual vs target score per exam

Chapter readiness score

Each chapter earns points as the student moves through Alcovia's rigour, so syllabus progress is measured by depth, not just "read it".

Stage	Proof	Points
1. Read the chapter	Marked in app	20
2. Notes made	Photo uploaded	20
3. Practice questions	Accuracy above 70%	20
4. Past paper, timed	Score logged	20
5. Exam ready	Accuracy above 80%	20

A subject's readiness = average across its chapters. An alert fires when a subject falls more than 15 points behind the quarter's target.

What the advisor can and can't do (not a subject expert)

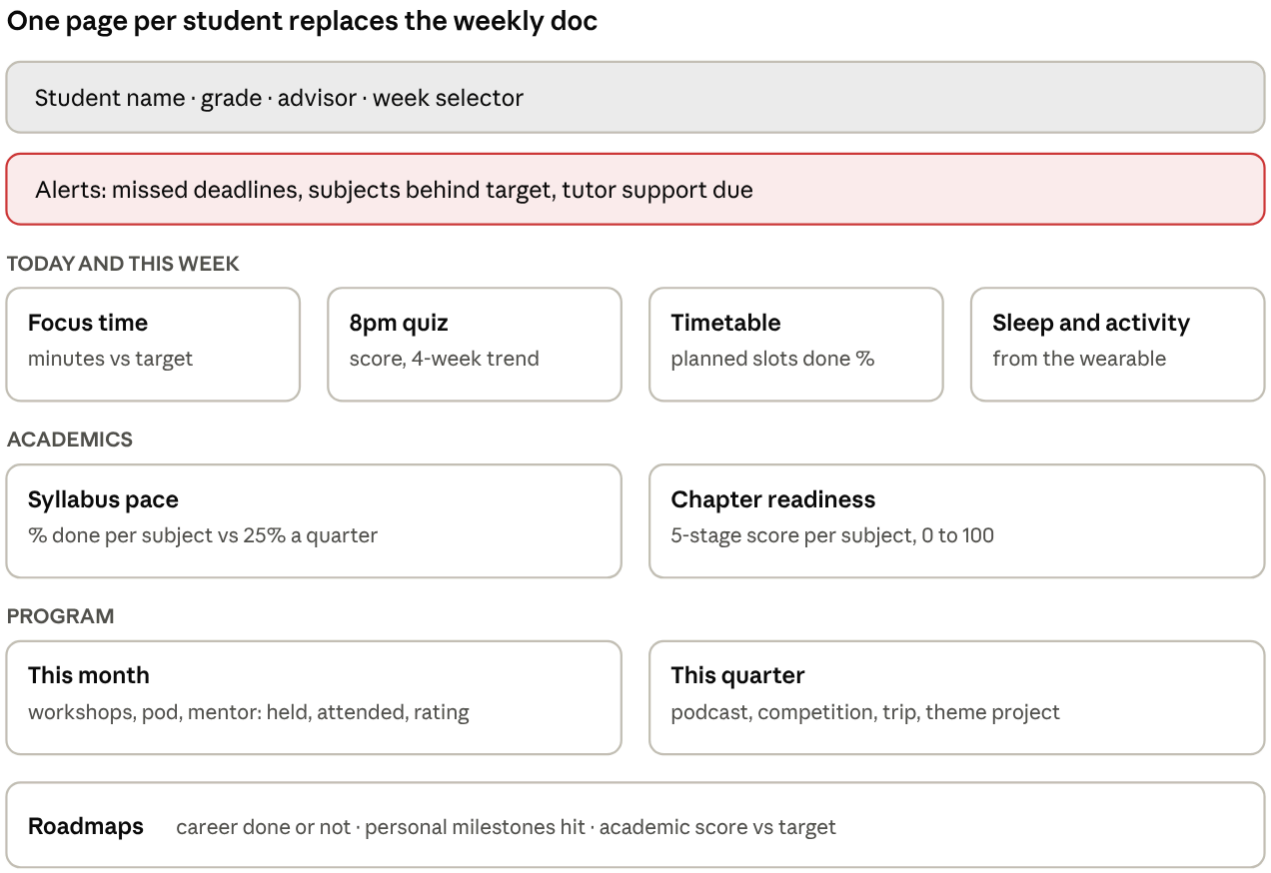
Advisors manage the **process**, not the content: set score targets from subject affinity and past results, build the timetable around tuition and extracurriculars, check that each chapter goes through all 5 stages, and arrange tutors or academy support when a subject falls more than the agreed percentage behind.

Advisor input KPIs. The dashboard also measures the advisor, because automations are only as accurate as the inputs behind them.

Advisor input	KPI	Target
Score targets agreed with student and parent	% of students with signed-off targets per subject and exam, with the logic written down	100% within the first month
Monthly chapter plan	% of subjects with a chapter plan for the month	100%
Quarterly rigour check	% of chapters reviewed against the 5 stages each quarter	100%
Student input cadence	% of students logging chapter stages at least 3 days a week	80% or more
Missed-deadline alerts	% of alerts with an action logged within 48 hours	90% or more
Tutor or academy support	Subjects behind by more than the agreed % get support arranged within 1 week	100%
Monthly sprints	Missed items consolidated into a sprint plan by the 5th of each month	Every month

Three views on one data set

- **Cohort view (founders):** averages across all students, with red flags (who is behind, which activity is under-delivered).
- **Student view (advisor):** one student on one page – day, week, month and quarter cards plus roadmaps. Replaces the weekly doc.
- **Parent summary:** a weekly WhatsApp message with 3 numbers and 1 highlight, sent by n8n.



Student view layout · alerts first, then day, academics, program, roadmaps

Alerts sit at the top so the weekly advisor call starts with what needs action; each card opens to the quantity, quality and performance KPIs above.

How data gets in with little manual work

Data	Source	How it arrives
Focus minutes, timetable, quiz scores, chapter stages	Alcovia app	Automatic, from app events
Apple Watch activity and sleep	Apple Health (HealthKit)	Read on the iPhone by the app; Apple has no cloud API
Samsung Watch and other Android wearables	Samsung Health via Health Connect	Read on the phone by the app
WHOOP	WHOOP developer API	Daily cloud sync (already connected)
Garmin	Garmin Connect Health API	Daily cloud sync, after Garmin approves the partner application

Data	Source	How it arrives
Session, workshop and mentor ratings	1-tap rating in the app after each session	Student taps once; n8n sends the reminder
Attendance and sessions held	Google Calendar and ClickUp	Synced by n8n
Soft-skill and workshop scores	Facilitator form (1–5 per skill)	Filled once per workshop on a phone
Exam scores and targets	Advisor enters once per exam in the app	The only regular manual entry

All sources write to one database (the app's backend). The dashboard reads from there, and n8n sends alerts to the advisor and parent when a deadline is missed.

Focus mode that locks the phone: what is feasible

The brief asks for a focus mode where no other app opens and only whitelisted numbers (parents) can call. This is the hardest feature here and needs a feasibility check before any promise.

Platform	Blocking other apps	Calls only from parents
iPhone	Possible with Apple's Screen Time API (needs Apple's Family Controls approval; the student or parent picks the apps to block)	The app cannot whitelist numbers itself; the student sets up an iOS Focus that allows only parents' calls, and the app guides them through it once
Android	Possible with an accessibility service or usage-access overlay (Google Play reviews this use closely)	Possible: with Do Not Disturb access the app can allow only starred contacts during a session

Plan: a 3-day check in week 1, a decision with founders, and the build in January if approved. Until then, focus time is tracked by the in-app timer, which already works.

Timeline (6 weeks)

Weeks	Me	Engineer	Process designer
1	Final KPI list and formulas; agree with advisors	Feasibility check on health data and focus mode	(on GTM tracker)

Weeks	Me	Engineer	Process designer
4–5	Design the 3 views; test with 2 advisors	(on app adoption)	Clean up ClickUp student folders into one structure; rating prompts; facilitator forms
6–7	Check every number against its source	HealthKit and Health Connect reading; WHOOP and Garmin sync; cohort and student views	Advisor training; parent summary via n8n
8	Launch; retire the weekly doc	Fixes	Weekly review moves to the dashboard
January	Decide on focus-mode build	Focus-mode build, if approved	Parent setup guide for whitelisted calls